

The Total Turing Test

In the early 1990s, cognitive scientist Steven Harnad added that the goal of the Turing Test should be expanded for modern AI systems to include two extra components. His position is that the Turing Test takes into account a “cold” scenario where interaction between the Interrogator and the intelligent agent is through a pure text scenario. But a modern AI, especially one in a robot, would need to pass tests of sight and physical interaction to truly be deemed “intelligent”. Russell and Norvig simplify these as computer vision and robotics.

- ▶ **Computer Vision:** In the Total Turing Test, the AI would also need to pass a test of sight. In this case, while the Interrogator can’t see A or B, he would be able to test their vision in some way to ascertain whether they’re human or machine. For an eventual intelligent robot, this would need to be a crucial element – and it would go on to test the knowledge representation aspect of the system a lot more thoroughly.
- ▶ **Robotics:** While machine learning is tested in the form of detecting new environments, the Turing Test doesn’t check the AI’s ability to manipulate these environments. How can a modern AI system – used so frequently to operate heavy machinery – be considered complete without the ability to interact with its environment in a way that most humans would?

Not An Exact Science

While it’s the most widely accepted form of testing artificial intelligence, the Turing Test can’t be called an exact science. It doesn’t give any denomination to intelligence nor any measurable capacity to pit one AI against another AI. To put it bluntly, the Turing Test is “all or nothing” in the sense that it can judge a machine to be artificially intelligent or not.

This obviously begs the question of what is intelligence. If a machine is more intelligent than any other animal on the planet, but less intelligent than a human being, is it still intelligent? And if a machine does pass the Turing Test and several others do the same later, then which is the “smartest” machine of them? The lack of a quantitative system in the Turing Test makes it an inexact scientific tool.

Still, currently, there aren’t any alternatives either. The closest anyone has come has been a team of scientists who are trying to come up with a “universal intelligence system”.